The Circuit Imprint: Loss-Surface Geometry Fingerprints the Grokking Transition

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Abstract

Grokking—where a network generalises long after memorising its training set—involves an algorithmic switch from a dense lookup table to a sparse Discrete Fourier Transform (DFT) circuit. Singular Learning Theory (SLT) predicts this switch leaves a geometric signature in the loss landscape: the Fourier solution should occupy a flatter, more degenerate minimum. But degeneracy itself is multi-faceted: a minimum can be *locally flat* (few constrained parameter directions) or *globally multiply represented* (many equivalent solutions scattered in weight space), and different estimators are sensitive to each. We study both simultaneously across grokking using the Hessian-estimated LLC (sensitive to local flatness) and the SGLD-estimated LLC (sensitive to global multiplicity), and find that these two aspects of the transition evolve very differently: local curvature collapses dramatically at the grokking transition, while global multiplicity remains roughly constant. The local signal is especially informative—the Hessian-estimated LLC settles near $19{\pm}2$ post-grokking (mean $\pm$ std, 5 seeds), consistent with ~ 12 active Fourier frequencies times two parameters each. A direct Fourier-amplitude analysis of the token embeddings confirms this independently, showing that geometry-based SLT probes can recover the effective complexity of a learned circuit without any circuit identification.

1. Introduction

Grokking (Power et al., 2022) compresses a rich learning phenomenon into a single training run. A transformer trained on modular arithmetic $(a{+}b) \bmod p$ rapidly fits its training examples, then enters a long plateau before abruptly generalising to the held-out pairs. From a mechanistic interpretability perspective this is well understood: the model switches from a dense, attention-based lookup table to a compact DFT circuit that uses only a small number of active Fourier frequencies (Nanda et al., 2023). The switch is algorithmic, not merely a refinement—the two solutions have qualitatively different structures.

Singular Learning Theory (SLT) (Watanabe, 2009) provides a complementary, geometry-based account of this same transition. Its central quantity—the Real Log Canonical Threshold (RLCT) λ—measures how *degenerate* a loss minimum is: a flatter, more symmetric minimum has a lower RLCT, a lower free energy, and a stronger implicit Bayesian prior. The sparse Fourier solution, with far fewer effective parameters than the dense lookup table, should be more singular in this sense. Weight decay amplifies this preference, penalising the high-norm lookup table faster than the compact Fourier circuit (Varma et al., 2023; Hoogland et al., 2024).

Two faces of degeneracy. What makes grokking an ideal SLT testbed is that the Fourier solution is degenerate in two distinct ways. First, *local flatness*: at any given Fourier solution the landscape is nearly flat in all directions except those tied to the active Fourier features—only ~ 24 effective parameter directions out of hundreds of thousands are constrained. Second, *global multiplicity*: many equivalent Fourier solutions exist, corresponding to different active frequency subsets related by DFT symmetries. Different estimators are sensitive to each: the Hessian measures local curvature at a single point; the SGLD chain can drift between equivalent solutions if its step size is large relative to local curvature, probing global multiplicity instead.

Questions and contributions. This paper asks: how does each geometric face of the transition evolve? Which carries the cleaner signal of the algorithmic switch? And can geometry alone recover what mechanistic interpretability finds?

1. We show that **local flatness** (Hessian-estimated LLC) collapses dramatically at the grokking transition and

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

directly fingerprints the Fourier circuit's intrinsic dimensionality—a result we term the **Circuit Imprint** (§4).

2. We show that **global multiplicity** (SGLD-estimated LLC) remains roughly constant across the transition, and explain why: a miscalibrated SGLD chain drifts between equivalent Fourier solutions rather than staying within one, probing the global count rather than the local geometry (§5).

3. We provide a direct cross-validation: Fourier-amplitude analysis of the token embeddings and Hessian curvature of the loss surface converge to the same effective dimensionality at the same training step with no shared information—demonstrating that geometry-based SLT probes can serve as a task-agnostic substitute for circuit identification (§4).

2. Background

Local Learning Coefficient. The RLCT λ appears in the free energy as $nF_n = nL_n + \lambda \log n + O(\log\log n)$ (Watanabe, 2013). A lower λ means a more degenerate minimum with a larger effective basin—SLT's rigorous account of implicit regularisation. The LLC $\hat{\lambda}$ estimates λ by running a localised SGLD chain near a trained weight w^*:

$$\hat{\lambda} \;=\; \beta n \, \mathbb{E}_{\mathrm{SGLD}}\big[L(w) - L(w^*)\big], \quad \beta = 1/\log n, \tag{1}$$

with a spring $\frac{\gamma}{2}\|w - w^*\|^2$ to keep the chain near w^*. The chain accumulates excess loss proportional to the volume of the basin it explores. If the step size is small relative to the local curvature, it stays within the local basin and measures local flatness. If it is large, it may drift toward other equivalent minima and measure global multiplicity instead. Calibration therefore determines which aspect of the RLCT the chain probes.

Hessian-estimated LLC. The Hessian $H = \nabla^2 L(w^*)$ encodes local curvature directly: large eigenvalues mark constrained directions; near-zero eigenvalues mark flat, unconstrained ones. Under the Gaussian approximation, $\lambda \approx \mathrm{rank}(H)/2$, motivating:

$$\hat{\lambda}_H \;=\; \frac{\mathrm{tr}(H)}{2\,\lambda_{\max}}, \tag{2}$$

which counts effective non-flat directions divided by two. We estimate $\mathrm{tr}(H)$ via the Hutchinson finite-difference estimator (Hutchinson, 1990) and $\lambda_{\max}$ via power iteration—both requiring only forward passes. Because $\hat{\lambda}_H$ is purely local it measures only local flatness, regardless of how many equivalent solutions exist elsewhere in weight space.

3. Setup

We train a one-layer transformer ($d_{\mathrm{model}} = 128$, 4 heads, $\sim$224k parameters) on $(a+b) \bmod 97$ (Power et al., 2022) with a 30% training split, AdamW ($\eta = 10^{-3}$, weight decay 1.0) (Kingma & Ba, 2015; Loshchilov & Hutter, 2019) for 5 000 steps. The high weight decay is essential: it penalises the high-norm lookup table faster than the sparse Fourier circuit, providing the energetic incentive for the transition (Varma et al., 2023). All results are averaged over 5 independent random seeds (seeds 0–4), with grokking occurring between steps 2 250 and 3 000 (mean ≈2 700). Figures show mean ± std across seeds.

At every 250 steps we compute: **(i)** SGLD-LLC via Equation (1) (200 steps, $\varepsilon = 3\times10^{-5}$, $\gamma = 100$); **(ii)** $\mathrm{tr}(H)$ and $\lambda_{\max}$ via Equation (2); and **(iii)** the token-embedding Fourier amplitude spectrum, $A_k = \left(\sum_d |\hat{W}_{k,d}|^2\right)^{1/2}/p$, where $\hat{W}$ is the DFT of W_E along the token axis.

4. Results

Three phases of grokking (Figure 1). Training unfolds in three well-separated phases. *Phase 1—Memorisation:* training loss falls to near zero while test loss rises above the random-chance baseline. The lookup-table solution fits every training example but actively suppresses generalisation. *Phase 2—Plateau:* both losses stagnate and the model appears stuck. *Phase 3—Grokking:* test loss drops sharply to near zero while training loss stays low—the Fourier algorithm has replaced the lookup table. This transition is invisible in training loss alone; the geometric probes we now describe detect it earlier and more cleanly.

Local and global geometry diverge at grokking (Figure 2). Figure 2 shows both probes across training. During memorisation both estimators are high—the lookup-table basin is complex and sharp. At the grokking transition the two measures sharply diverge: $\hat{\lambda}_H$ collapses to a much lower value and stabilises, confirming that the Fourier basin is dramatically flatter. The SGLD-LLC barely changes. Post-grokking the two sit roughly two orders of magnitude apart on a log scale.

This divergence is informative, not contradictory. The Hessian reports on the local geometry of whichever Fourier solution the model landed in. The SGLD chain, with a step size large relative to the post-grokking curvature, drifts between many equivalent Fourier solutions and reports their global count. Both signals are real; they characterise different geometric aspects of the same transition.

The right panel shows the ratio of the two over time: modest during memorisation, jumping sharply at grokking onset and remaining elevated. This ratio is a geometry-based

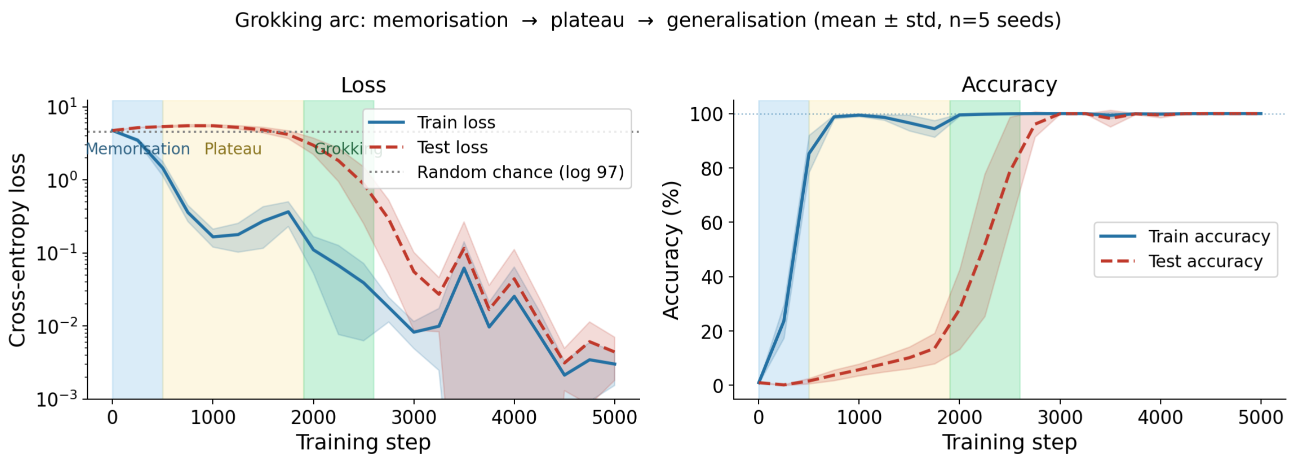

Figure 1. **Three-phase grokking arc** (mean $\pm$ std, 5 seeds). The three phases—memorisation (blue), plateau (orange), and grokking transition (green)—are annotated consistently across all figures. Training accuracy reaches 100% early while test accuracy stays near random chance until the abrupt transition (mean step $\approx$2 700). Test loss peaks above the random-chance baseline during the plateau, confirming the lookup-table solution actively suppresses generalisation before the algorithmic switch occurs. The shaded bands confirm the three-phase structure is robust across random seeds.

transition detector that requires no labelled test set and fires at precisely the right step.

The local signal is particularly informative about the circuit. Post-grokking, $\hat{\lambda}_H$ settles near 19 ± 2 (mean $\pm$ std across 5 seeds)—consistent with ~ 12 active Fourier frequencies, each with two free parameters, giving $12 \times 2 = 24$ effective dimensions. The Hessian has, without any Fourier analysis, independently recovered the circuit's intrinsic dimensionality. We call this the **Circuit Imprint**: the loss-surface curvature carries a quantitative fingerprint of the learned algorithm.

Cross-validating the Circuit Imprint (Figure 3). The Circuit Imprint makes a testable prediction: if $\hat{\lambda}_H \approx 19$ is counting active Fourier dimensions, a direct frequency analysis of the token embeddings should find ~ 12 active frequencies activating at the same step.

The left panel shows the Fourier spectrum of W_E as a heatmap. Early in training and through the plateau all frequencies have low, equal amplitude— random noise with no preferred structure. At the grokking transition, a small set of frequencies brightens sharply while the rest stay dim. They activate together, at the same step as the accuracy jump.

The right panel overlays the active frequency count (orange) with $\hat{\lambda}_H$ (teal). Both signals transition at the same training step and converge to consistent values post-grokking. Crucially, they share no information: the spectral analysis knows nothing about the loss landscape; the Hessian knows nothing about the circuit structure. Their simultaneous convergence confirms that loss-surface curvature encodes the circuit's intrinsic dimensionality in a precise, recoverable form—without any task-specific analysis.

5. Discussion

Understanding the SGLD-LLC signal. The SGLD-LLC staying high post-grokking is not a failure—it is measuring something real. The Fourier solution has high *global multiplicity*: many equivalent solutions exist, corresponding to different active frequency subsets and phase rotations, all related by the symmetries of the DFT. With the step size used here, the SGLD chain drifts across the vast flat subspace and moves between these equivalent solutions, accumulating excess loss and yielding a high $\hat{\lambda}$. A step size tuned to the post-grokking curvature would confine the chain to a single solution and bring the SGLD-LLC in line with $\hat{\lambda}_H$. The two estimators are not in conflict—they are instruments calibrated to different geometric scales.

The Circuit Imprint as a task-agnostic probe. Mechanistic interpretability identifies the Fourier circuit through task-specific projections that require prior knowledge of the task's structure (Nanda et al., 2023). The Circuit Imprint shows that $\hat{\lambda}_H$ arrives at the same answer via a task-agnostic route: the loss surface carries a curvature signature of the active circuit, readable without knowing which frequencies to look for. The two approaches are complementary—mechanistic interpretability identifies *which* components are active; SLT geometry quantifies *how many effective parameters* the solution uses. Where circuit identification is costly or task-specific, $\hat{\lambda}_H$ may serve as a practical substitute.

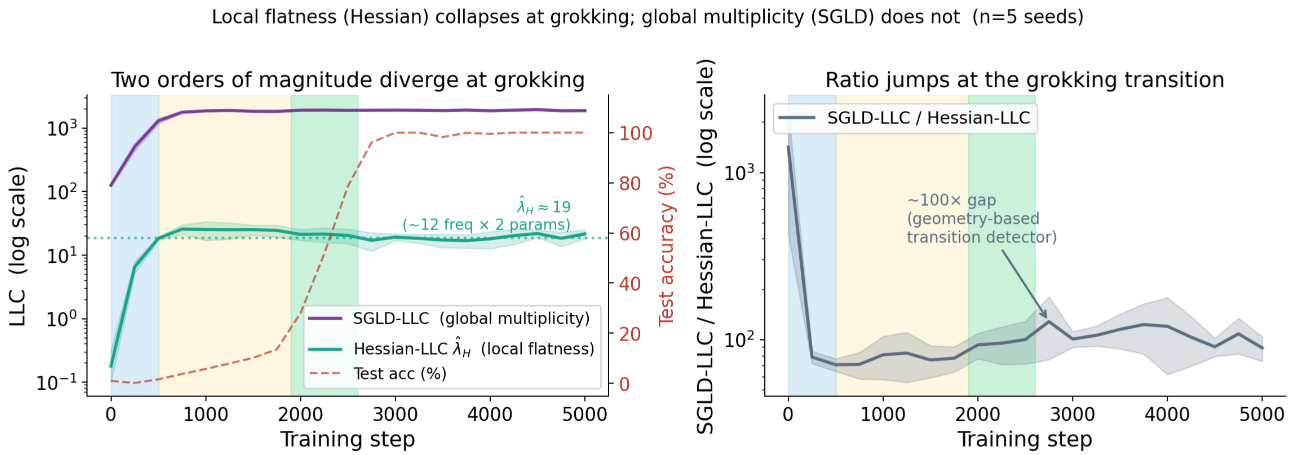

Figure 2. **Local flatness and global multiplicity diverge at grokking** (mean ± std, 5 seeds). *Left:* SGLD-LLC (purple, sensitive to global multiplicity) and Hessian-estimated LLC $\hat{\lambda}_H$ (teal, sensitive to local flatness) on a shared log axis, with test accuracy (green dots). Both are high during memorisation; at the grokking transition $\hat{\lambda}_H$ collapses while SGLD-LLC stays elevated—two orders of magnitude apart post-grokking. The value $\hat{\lambda}_H = 19 \pm 2$ matches the intrinsic dimensionality of the Fourier circuit (~ 12 frequencies $\times$ 2 parameters). *Right:* Ratio of the two estimators over time. The ratio jumps at the grokking transition and remains elevated—a geometry-based phase detector that requires no labelled test set. Shaded bands show variability across seeds; the dissociation is consistent across all runs.

Relation to the competing-basins picture. Recent work (Author(s) of arXiv:2603.01192, 2026) frames grokking as a phase transition between two geometrically distinct basins, tracking the LLC as a continuous transition detector. Our results refine this picture. The SGLD-LLC alone does not cleanly separate the two basins: it rises during memorisation but does not drop post-grokking. Local curvature, by contrast, begins collapsing *during* the grokking transition itself, providing an earlier signal. When both probes are available, the trajectory in (Hessian curvature, LLC) space traces a clean path between two well-separated clusters—one for the memorisation basin and one for the Fourier basin—with the curvature axis carrying most of the discriminating power.

6. Conclusion

Grokking turns out to expose two distinct faces of degeneracy in the loss landscape. The Fourier basin is locally flat: almost all parameter directions are unconstrained, and the few that carry curvature quantitatively fingerprint the active Fourier circuit—the Circuit Imprint. It is also globally multiple: many equivalent Fourier solutions exist, and a miscalibrated SGLD chain explores them rather than characterising any single one.

These two faces are not in conflict; they reflect genuinely different geometric properties of the solution. But they have different practical implications. Local flatness, probed by the Hessian-estimated LLC, tracks the algorithmic switch cleanly, recovers the circuit's intrinsic dimensionality without circuit identification, and is confirmed independently by Fourier-amplitude analysis of the token embeddings. Global multiplicity, probed by the SGLD-LLC, reflects the symmetry group of the Fourier solution—real and theoretically interesting, but insensitive to the transition itself at standard step sizes.

Together, these results suggest that decomposing the RLCT into its local and global components is a productive lens for studying phase transitions in trained models—and that loss-landscape curvature can serve as a task-agnostic proxy for circuit complexity in settings where mechanistic identification is costly or unavailable.

Impact Statement

This paper advances the theoretical understanding of generalisation in transformer models. There are no ethical concerns or negative societal consequences associated with this work.

Acknowledgements

[To be added in camera-ready version.]

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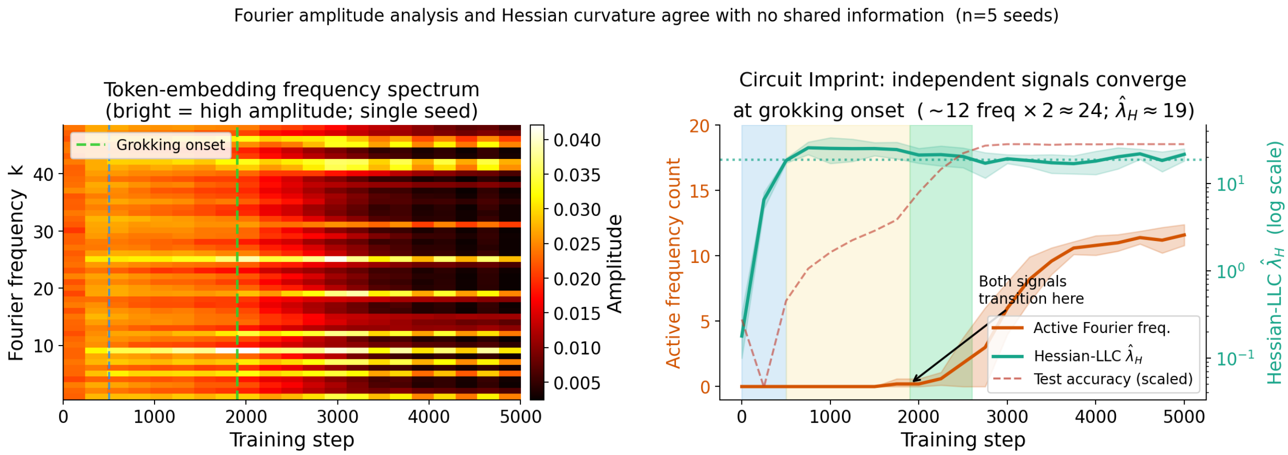

Figure 3. **Circuit Imprint: loss-geometry independently recovers the circuit's dimensionality.** *Left:* Token-embedding Fourier spectrum (single representative seed; bright = high amplitude). Through the memorisation phase all frequencies are dim and equal; at the grokking transition a small set abruptly activates (vertical green dashed line) while others remain near zero. *Right:* Active frequency count (orange, $\sim$12 post-grokking) and $\hat{\lambda}_H$ (teal, 19 ± 2 post-grokking) both transition at the same training step and reach consistent values post-grokking. The two signals are derived from entirely different objects (embedding spectrum vs. loss-surface curvature) and share no information, yet converge simultaneously across all seeds.

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